PORTFOLIO

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AT A GLANCE

Master's in Physics + Engineering degree + PhD @ Strasbourg, Tokyo, Munich

2 years of diplomatic experience

- 20+ high-level scientific diplomacy events organised
- Drafted notes on AI, Quantum, Space, Biotech, Innovation
- Built networks in the political, diplomatic, academic and industrial spheres

5 years of research experience

- 2 high-impact peer-reviewed publications (*Nature Communications*, *iScience*)
- 4 talks, 2 posters in international conferences (France, Germany, Japan)
- Skills: experimental (bio)physics, project management, programming and data-science (Python)

AS A DIPLOMAT (2024-2026)



Moderating for the 10th anniversary of the Paris Agreement on Climate in Bern

As **Scientific Officer** for the **French Embassy to Switzerland**, I had the chance of organising **20+ diplomatic events**, related to science, research, higher education, and great societal challenges.

From drafting **concept notes**, planning the **logistics** of the event, **liaising with speakers**, **communicating**, and **moderating** panels, I covered the entire organisation chain.

ENVIRONMENTAL CHALLENGES

In cooperation with the Swiss Federal Department of Foreign Affairs, we organised **outreach conferences** to explain the role of scientists and diplomats working to preserve the planet.

- From the Paris Climate Agreement to the One Health Summit
- From the 3rd U.N. Conference on Oceans, to the Arctic with the CASCADES Expedition



Moderating a panel with IPCC/GIEC chairs Profs. Seneviratne and Stocker, and economist Dr. Fabre

INDUSTRIAL AND SCIENTIFIC CHALLENGES



Left: *Moderating a panel on* ***supply chain resilience*** *with Framatome AG Director Mr. Felt, Paires CEO Mr. Saudin, and Swan Isotopen CEO Mr. Roncolato*

We brought together audiences of leading academic and industrial actors to exchange with decisionmakers on crucial **technological challenges** in the fields of :

- **Artificial Intelligence**
- **Quantum**
- **Nuclear Energy**
- **Space**

(GEO)POLITICAL CHALLENGES

We also explored burning (geo)political questions with experts from governments, international organisations, and think tanks such as:

- How does **A.I. amplifies disinformation**, and can it help mitigating it?
- How can **Europe** adapt to the new forms of **hybrid warfare**?
- How should organisations cooperate to prevent the next **pandemic**?



*Moderating a panel on **scaling-up safely the use of nuclear medicine**, with representatives of BAG/OFSP, and of the French and Swiss Nuclear Medicine societies*

PROMOTING COOPERATION



Lastly, I got a chance to contribute to the reinforcement of the scientific bonds that exist between France and Switzerland by:

- **Connecting** training/research/innovation **ecosystems** in both countries.
- **Administering**, the **funding programs** that enable those cooperations.
- **Exchanging** with the **students**, keys to fostering this relationship.

SOME NOTABLE EVENTS ORGANISED

Name	Date & Place	Notable partners & speakers
The CASCADES Expedition: International Scientific Cooperation for Arctic Research	September 18, 2026 @ Universität Bern | Link	**H.E. Balz Albplanalp** - Chief of the Prosperity and Sustainability division of EDA/DFAE, **H.E. Jean-Paul Lemieux** - Canadian Ambassador to Switzerland, Swiss Polar Institute
Nuclear Medicine: Therapeutic advances / the Role(s) of public actors / Industrial challenges	September 8, 2026 @ Résidence de France à Berne | Link	**Prof. Prior** - Swiss Society of Nuclear Medicine President, **Prof. Courbon** - French Society of Nuclear Medicine President,

Name	Date & Place	Notable partners & speakers
		Alexandre Felt - Director of Framatome AG, BAG/OFSP
Celebrating the alumni of French Higher Education in Switzerland	May 27, 2026 @ Grande Orangerie de Berne | Link	**Col. Yves Fournier** - Chief of Staff for the Swiss Air Forces, **Frank Rijnberg** - President of ESN Switzerland
Exploring the Universe, Transforming Medicine	May 4, 2026 @ Universität Bern | Link	**Dr. Claudie Haigneré**, former French Minister, and first French woman in space

Name	Date & Place	Notable partners & speakers
5th Kornhaus Dialog - European security and hybrid threats	March 26, 2026 @ Welle 7, Bern, Link	Embassy of Austria, foraus
One Health - From animals to humans: adressing zoonotic diseases	March 17, 2026 @ Haus der Universität Bern | Link	EDA/DFAE, WHO, IRD, INRAE
Handling current and future nuclear waste: needs, solutions, perspectives	October 30, 2025 @ Résidence de France à Berne | Link	**Benoît Revaz** - Director of BFE/OFEN, I2EN, Nuklearforum, Andra

Name	Date & Place	Notable partners & speakers
Protecting and preserving the oceans	May 27, 2025 @ Universität Bern | Link	**H.E. Alexandra Baumann** - Chief of the Prosperity and Sustainability division of EDA/DFAE
4th Kornhaus Dialog - Artificial intelligence and countering disinformation	February 20, 2025 @ Kornhaus Forum, Bern | Link	Embassy of Austria, foraus
Celebrating the French community of ETH Zurich	November 19, 2025 @ ETH Zürich	**Prof. Joël Mesot** - President of ETHZ

Name	Date & Place	Notable partners & speakers
10 years after the Paris Agreement, the fight continues: between aspirations and reality	November 10, 2025 @ Universität Bern | Link	**H.E. Alexandra Baumann** - Chief of the Prosperity and Sustainability division of EDA/DFAE, **Laurence Tubiana** - Key architect of the Paris Agreement, **Prof. Sonia Seneviratne** & **Prof. Thomas Stocker** - Chairs of IPCC/GIEC panels
Franco-Swiss Days for Higher Education & Research	May 22-23, 2025 @ Universität Basel | Link	**40 French and Swiss institutions**: universities, ministries, startup incubators

Name	Date & Place	Notable partners & speakers
French Nuclear Industry Day	March 14, 2025 @ Paul Scherrer Institut | Link	PSI, CEA, Framatome, newcleo
10 years of Franco-Swiss cooperation: ANR & SNSF	December 10, 2024 @ Résidence de France à Berne | Link	**Dr. Martina Hirayama** - SBFI/SEFRI State Secretary, **Dr. Claire Giry** - ANR Director-President, **Dr. Matthias Egger** - SNF/FNS Research Council President

AS A RESEARCHER (2019-2024)

ABOUT MY RESEARCH

Pancreatic cancer is one the world's **deadliest cancer**, where patients remain asymptomatic until a diagnosis at advanced stages. Despite massive progresses in the understanding of the cancer's biology, ***therapeutic gains have not materialised***, as **current in vitro models cannot accurately model** the development of cancerous lesions.

Our research team developed an in vitro model of tumours that **faithfully recreate the architecture** of the lesions and that allows **real-time observation of their dynamics**.
This represents a crucial step in understanding how cancer forms and progresses, and opens perspectives for the **development and the testing of new therapies**.

PUBLICATIONS

- **“Spatiotemporal dynamics of self-organized branching in pancreas-derived organoids”** Nature Communications 13, 5219 (2022) [Link]
  - **S. Randriamanantsoa*** , A. Papargyriou*, H. C. Maurer, K. Peschke, M. Schuster, G. Zecchin, K. Steiger, R. Öllinger, D. Saur, C. Scheel, R. Rad, E. Hannezo, M. Reichert & A. R. Bausch.
- **“Coexisting mechanisms of luminogenesis in pancreatic cancer-derived organoids”** iScience (2024) [Link]
  - **S. Randriamanantsoa***, M. Raich*, D. Saur, M.Reichert, A. R. Bausch

INTERNATIONAL CONFERENCES

- Selected talk - **ISMC 2023 7th International Soft Matter Conference**, Osaka, Japan - 09/2023 [Link]
- Invited talk - **The 6th International Conference on Physics and Biological Systems**, ENS Paris-Saclay, France - 06/2022 [Link]
- Selected talk - **Microphysio 2022 conference "Microphysiological systems from organoids to organ-on-chips"**, Cargèse, France - 04/2022 [Link]
- Selected talk - **AQV Days 2022 - Quantitative Approaches to Living Systems**, Institut Jacques Monod, Paris, France - 01/2022 [Link]

- Selected Poster - **Centuri 2019 Scientific Meeting (CSM 2019)**, Cargèse, France - 10/2019 [Link]
- Selected Poster - **SFB 863 “Forces in Biomolecular Systems”**, Schloss Ringberg, Germany - 07/2019

CODE & DATASCIENCE

Using **Python and Jupyter notebooks**, I wrote code to load, clean, process and visualise experimental data, with the help of the pandas, scipy, matplotlib, and seaborn.

You can visualise a sample of the code and the output of the notebooks on: github.com/samrandria/sample-academia-code

Full code and datasets for the academic papers can be found at:

- Spatiotemporal dynamics of self-organized branching in pancreas-derived organoids - Zenodo
- Coexisting mechanisms of luminogenesis in pancreatic cancer-derived organoids - Zenodo

DISTINCTIONS

June 2023

Selected as
“one of the 600 most qualified young scientists worldwide”

– to attend the 72nd Lindau Nobel Laureate Meeting, and exchange with more than 40 Nobel laureates.

“The Lindau Nobel Laureate Meetings are annual scientific conferences held in Lindau, Bavaria, Germany, since 1951.

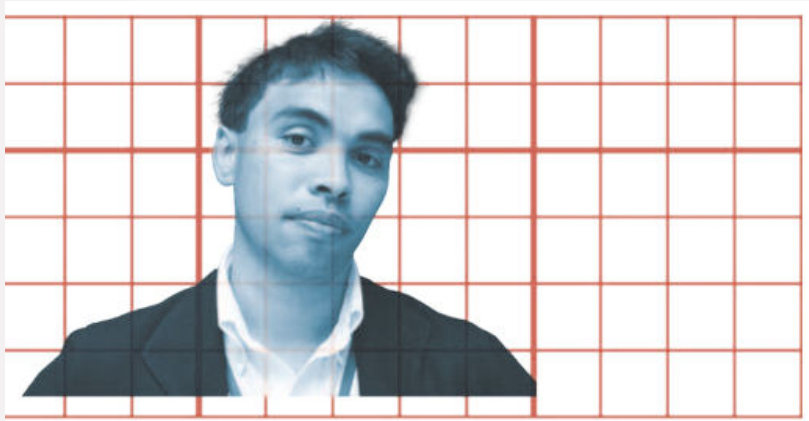
Their aim is to bring together **Nobel laureates** and the **world’s most talented young scientists** to ***foster scientific exchange*** between different generations, cultures and disciplines.“ [Link]

For this occasion, I was interviewed by the **Frankfurter Allgemeine Zeitung** to talk about my experience there as a young researcher and “believing in the way forward” in science. - 07/2023 [Link]

Zukunftslabor Lindau

SEITE B 4 · SAMSTAG, 8. JULI 2023 · NR. 156

FRANKFURTER ALLGEMEINE ZEITUNG



An den Weg vorwärts glauben

Samuel Randriamanantsoa arbeitet in München in einem interdisziplinären Projekt an Krebs-Organoiden

EXTERNAL SEMINAR TALKS

- Selected talk - **SFB 1032 "Nano Agents" Final Workshop**, Schloss Lautrach, Germany - 06/2024
- Invited talk - **ibiTea webinar**, ibidi GmbH, (online) - 06/2023 [Link]
- **Munich Institute of Bio Engineering (MIBE)**, Garching Germany - 05/2023 [Link]
- Seminar for the group of **Prof. Yanlan MAO** - University College London, United Kingdom - 11/2022
- **SFB 1032 "Nano Agents" Annual Workshop** - Munich, Germany - 10/2022 [Link]
- Seminar for the **Center of Protein Assembly** - Garching, Germany - 2022
- Seminar for the group of **Prof. Tetsuya HIRAIWA** - Mechanobiology Institute, Singapore - 08/2022

Additional talks

I also occasionally talk about broader issues in science during internal seminars:

“ChatGPT and Large Langage Models in scientific writing - Before using Generative AI in your next manuscript” - E22/E27 Winter School 2024

The emergence of publicly-available large language models (LLMs) has garnered considerable attention, and some in the scientific community have been contemplating using those tools to write their next papers. This talk explores the issues associated to this idea, notably the questions of copyright and licencing, of plagiarism, of confidentiality, and of responsibility that accompany the use of current LLMs. I introduce examples and litigation cases, often outside of academia, to illustrate those issues.

“Statistics done wrong” - E22/E27 Winter School 2023

Scientists love to use the words “statistically significant” or manipulate “p-values” to draw conclusions (!) about hypotheses. In most of the cases, this is done in a terribly wrong fashion, as has repeatedly been reported (see here, here, here or here). Yet, the quest for 3 (or 4, or 5) stars of “significance” to put above barplots continues **plaguing research**. The ambition of this talk is to illustrate how bad this can get, to get the audience to **hopefully re-consider before carelessly using statistics** in their next paper.

TEACHING AND SUPERVISION

As part of my work at TUM, I have had the privilege of supervising great students at the Bachelor and Master levels for their thesis.

- **Sangwon Lee** - “Drastic fluid uptake by pancreatic cancer organoids: tip swelling and pearling” - Max Planck School Matter to Life, University of Göttingen - Master 2023
- **Eric Waschkau** - “Scaffold-guided morphogenesis of pancreatic ductal adenocarcinoma cells in a microfluidic chip” Technische Universität München - Master 2021
- **Matús Novák** - “The Role of the E-Cadherin During Lumen Formation in Pancreatic Ductal Adenocarcinoma Organoids” - Technische Universität München - Bachelor, 2021

- **Giulia Zecchin** - “Branch extension in pancreatic organoids” - Technische Universität München - Bachelor, 2020
- **Anna Smettan** - “Retroviral transduction of a murine pancreatic cancer cell line for organoid growth” - Technische Universität München - Bachelor, 2020
- **Guillermo Prol Castello** - “Origin of dynamic nematic order in epithelial cell layers” - Universitat Politècnica de Catalunya - Bachelor, 2019

I also taught at the Bachelor level:

- Practical classes in **Biophysics** and **Thermodynamics**
- Tutorial classes in **Physics**, **Optics** and **Electromagnetism**

CREDITS

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